01

Adaptive Water Extraction

with Q Learning under Climate Stress

An agent based model in Python

TEAM

Group Members

Five contributors. One model. One simulation.

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AGENDA

Contents

A guided tour of the model and its findings

01 . CONTEXT

The Problem

Tragedy of the Commons meets a changing climate

- Twenty farmers share one river. Each season every farmer decides how much water to extract.
- If everyone extracts heavily the resource collapses and everyone loses next year. Hardin (1968) called this the Tragedy of the Commons.
- Pakistan's Indus Basin sustains over 90 percent of the country's agriculture and is one of the world's most water stressed basins.
- By 2030 nearly half the global population will live in regions of high water stress (UN World Water Development Report).
- Real farmers do not follow fixed rules. They learn, observe neighbors, and adapt. Existing agent based models do not capture this learning.

01 . FOCUS

Research Question

PRIMARY QUESTION

How does the discount factor gamma in Q learning agents affect the emergence of sustainable water extraction strategies under varying climate stress scenarios?

SUB QUESTIONS

- Can Q learning agents discover cooperative strategies without external enforcement?
- How do learned strategies break down under sudden drought shocks, and how fast do agents re adapt?
- At what monitoring intensity does enforcement become unnecessary?

How the Model Works

Three stage step. Each agent learns from experience.

- Each irrigator chooses one of five extraction levels (0, 25, 50, 75, 100 percent of max).
- State observed by the agent is a tuple. Water level bin, own previous action, neighbors average action, climate stress flag.
- Reward equals log utility of water extracted minus a sustainability penalty minus any fine.
- Shared resource follows logistic regeneration with a baseline inflow modulated by climate factor.
- Monitor agent catches over extractors with probability p detect and issues a fine.
- Social network is a random k regular graph. Each farmer observes k neighbors.
- Implemented in Python 3.12 with Mesa 2.3.4. About 700 lines of clear code.

02 . MATH

The Q Learning Update Rule

Watkins and Dayan 1992. Tabular and interpretable.

s, a) = Q (s, a) + alpha [r + gamma max Q (s', a') minus Q (s,

alpha is the learning rate. gamma is the discount factor. s' is the post regen state.

- Exploration uses epsilon greedy with exponential decay. Epsilon starts at 0.30 and decays at rate 0.010.
- State space is small. About 150 unique state action pairs fit comfortably in a tabular Q table.
- Action 2 equals each agent's fair share of total sustainable yield. Auto scaled with N.
- Heterogeneity. Each agent has its own RNG and its own initial Q table noise.

03 . SCENARIOS

Climate Scenarios

Four climate factor profiles test resilience

Scenario	Climate factor profile
Stable	c_f = 1.0 forever. Baseline. No climate change.
Gradual	c_f declines linearly from 1.0 to 0.5 over 400 steps.
Shock	c_f = 1.0 normally. Drops to 0.3 between steps 200 and 300.
Stochastic	c_f is gaussian noise around 0.85, clipped to between 0.3 and 1.0.

c_f multiplies BOTH the logistic regeneration term AND the baseline inflow.

Lower c_f means less water replenished each season.

03 . EXPERIMENTS

Experimental Design

Three batch experiments totaling 340 runs

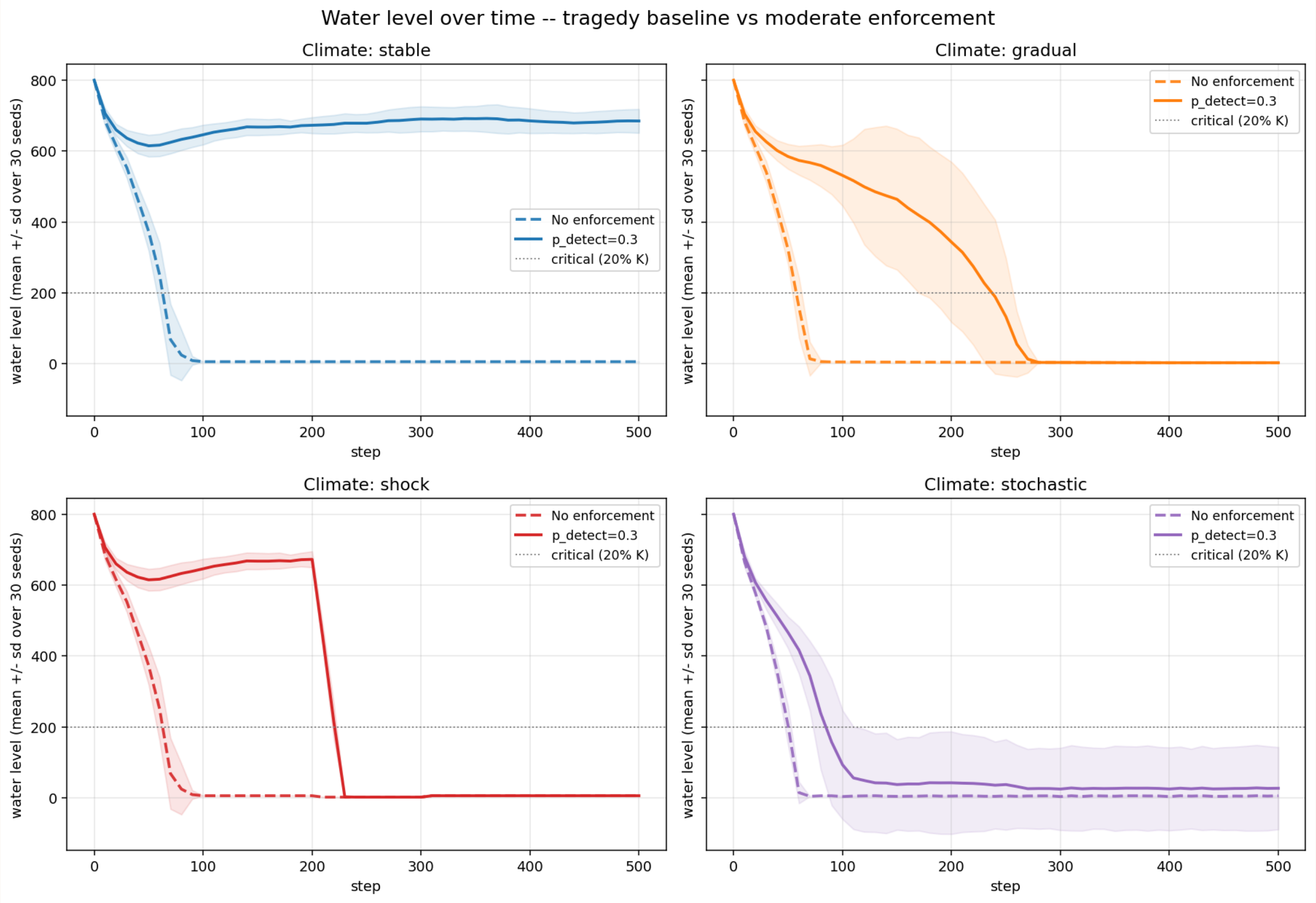
Experiment	Variables	Runs
A. Tragedy baseline	4 climates x 30 seeds. No monitor.	120
B. With enforcement	4 climates x 30 seeds. p detect = 0.3	120
C. Gamma sweep	stable and shock x 5 gamma values x 10 seeds	100

- Each run is 500 simulation steps. About 125 seasons if one step is one quarter year.
- Sensitivity analysis varies alpha, gamma, epsilon, regeneration rate, p detect, N by plus or minus 20 percent.
- Validation. Three sanity checks against Hardin 1968, Perolat 2017, and Indus Basin 2022.

— 04 . RESULT 1

Tragedy Baseline

Without enforcement every climate collapses



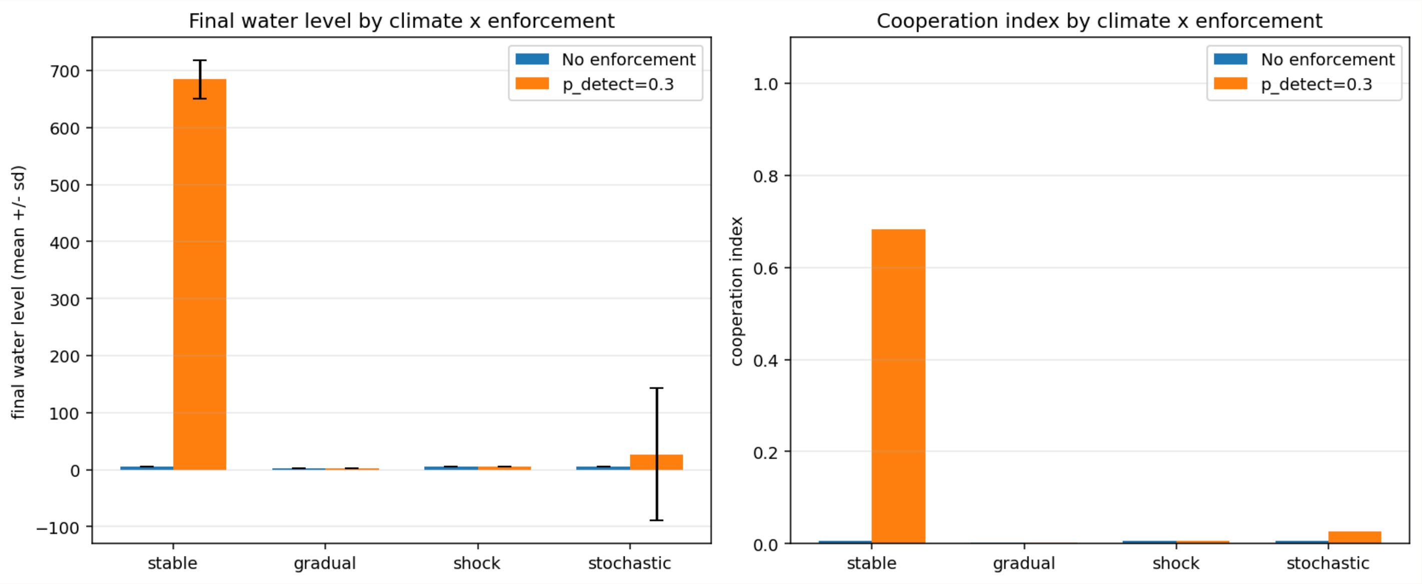
WHAT IT MEANS

- All four scenarios collapse to roughly 5 units of water within 80 steps.
- Mean payoff falls to between minus 755 and minus 804 per agent.
- Standard deviation across 30 seeds is effectively zero. Tragedy is deterministic.
- Reproduces Hardin (1968) at the population level.

05 . RESULT 2

Enforcement Saves the Stable Climate

Moderate monitoring flips the dynamics



WHAT IT MEANS

- Stable scenario. Water rises from 5 to 685. A 125 times improvement.
- Mean payoff turns positive. Minus 756 becomes plus 222 per agent.
- Eighty six percent of agents choose cooperative actions.
- Three other scenarios still collapse. See next slide for why.

06 . KEY FINDING

Why Enforcement Fails Under Stress

The fixed quota policy breaks when climate changes

Even p detect = 0.9 cannot save the drought shock scenario.

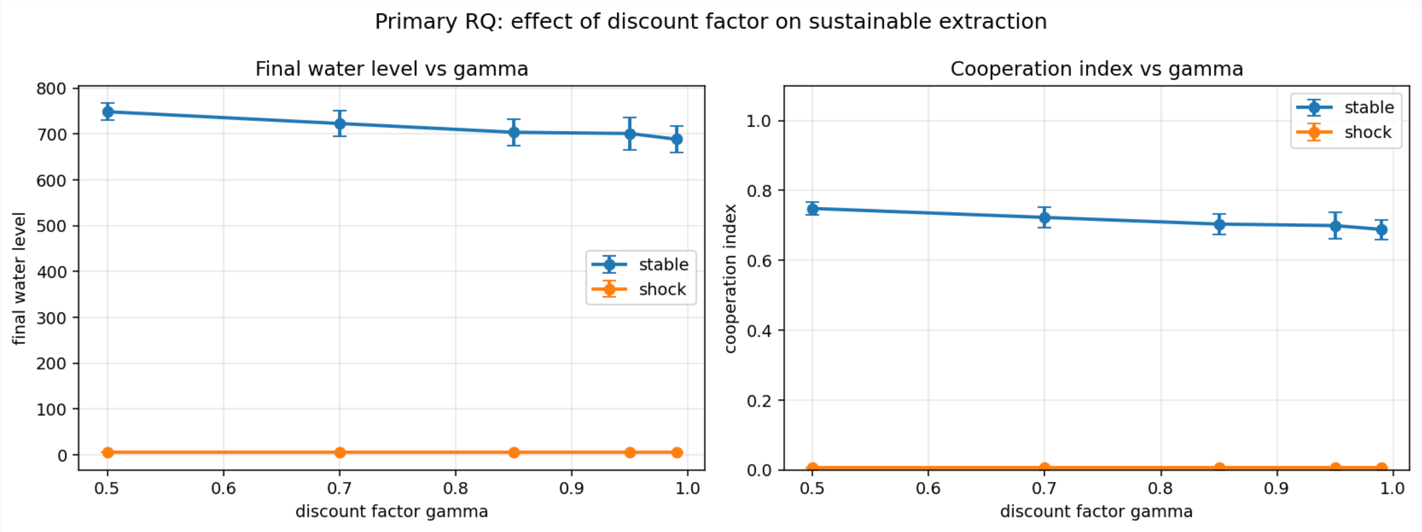
p detect	stable. final water	shock. final water
0.3	700	5.6 (collapsed)
0.6	777	5.6 (collapsed)
0.9	787	5.6 (collapsed)

WHY IT HAPPENS (structural, not a tuning issue)

- Agents' five action grid is calibrated to the baseline climate where c f = 1.0
- When c f drops to 0.3 during the shock, even the cooperative fair share action is over extraction.
- Sixty eight percent of agents are STILL choosing cooperative actions during failure. They are not selfish.
- Implication. Real regulators need adaptive (climate aware) quotas. Fixed quotas are not enough.

Effect of the Discount Factor

Gamma sweep over stable and shock scenarios

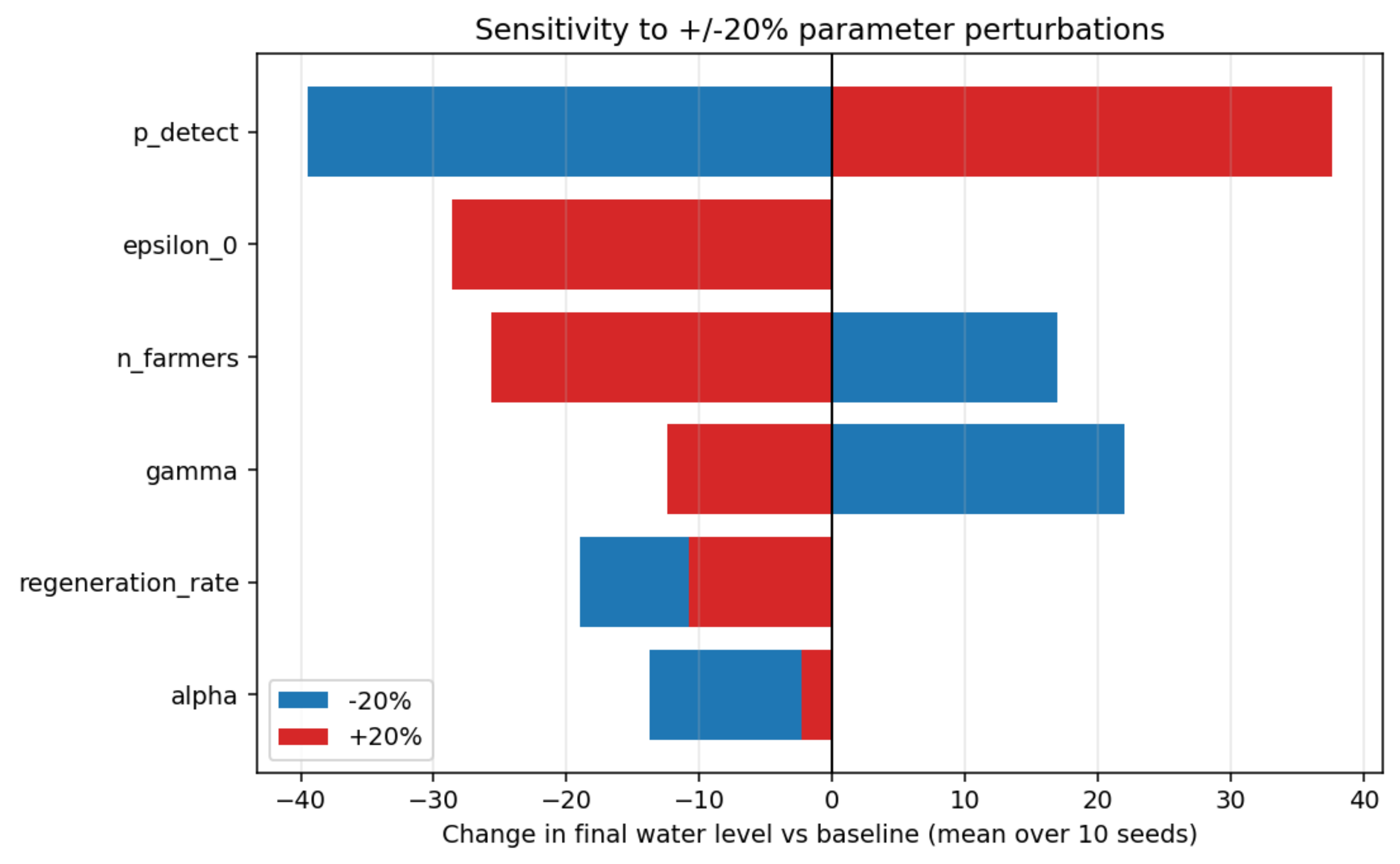


WHAT IT MEANS

- Stable scenario. Final water mildly DECREASES with gamma. From 748 down to 688.
- Shock scenario. Final water is identical at 5.55 across all gamma values.
- Surprising direction. Lower gamma (more myopic) is slightly better in stable climate.
- Refines the answer. Gamma is not the resilience knob. Action space matters more.

Sensitivity Analysis

Plus or minus 20 percent on six parameters



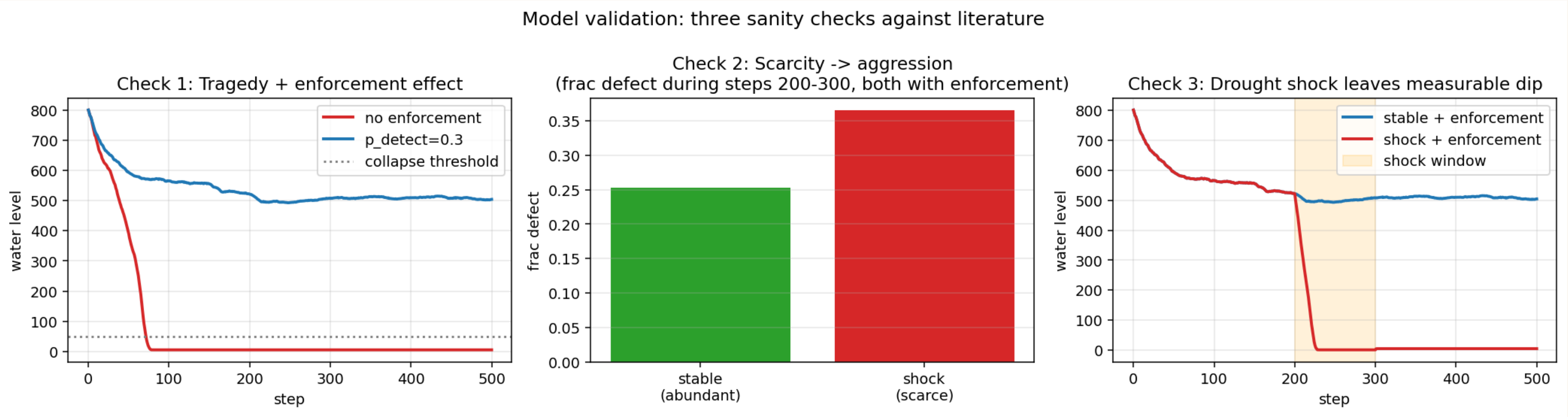
WHAT IT MEANS

- p detect dominates. Plus or minus 38 units of final water.
- Epsilon zero and alpha are non monotonic. Baseline is near a local optimum.
- Gamma effect is moderate. Regeneration rate is small.
- Validates that p detect = 0.3 is a fair choice for the headline runs.

Validation Against Literature

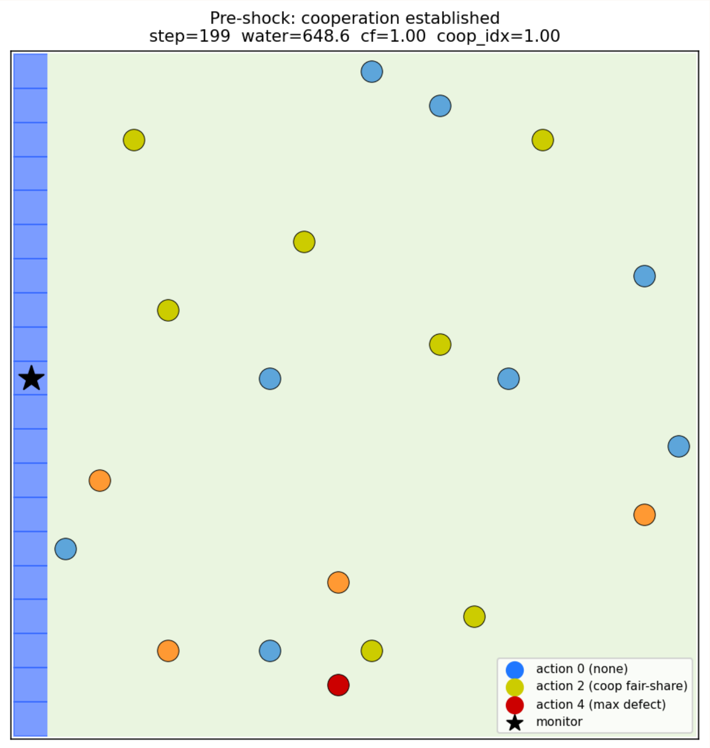
Three sanity checks. All three pass.

#	Source	Test	Result
1	Hardin 1968	No enforcement should collapse the resource. With enforcement under stable climate it should sustain.	PASS 5.6 vs 504
2	Perolat et al 2017	Scarcity should drive aggression. Compare frac defect during steps 200 to 300 under stable vs shock with enforcement.	PASS 25.3 percent vs 36.5 percent
3	Indus Basin 2022	A 70 percent drop in c f should leave a measurable dip in water level at step 300.	PASS 507 vs 1.5

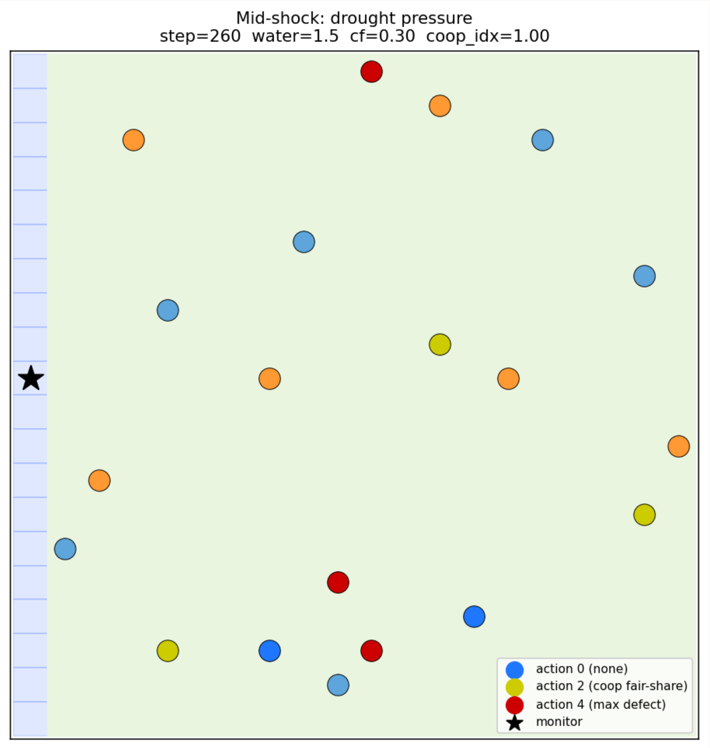


Emergent Spatial Dynamics

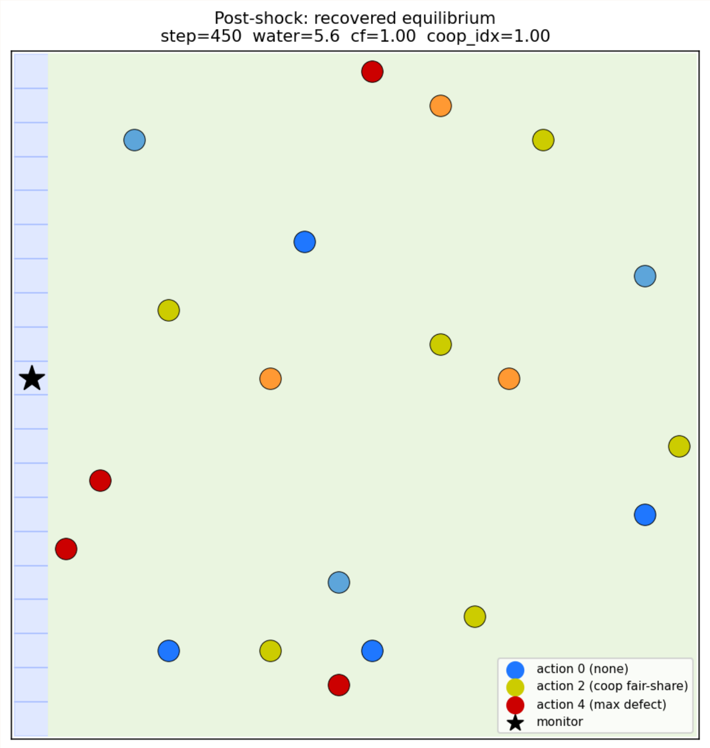
Three snapshots from one shock scenario run



Step 199. Pre shock cooperation



Step 260. Mid shock pressure



Step 450. Post shock equilibrium

Conclusions and Future Work

What we learned. What comes next.

FINDINGS

- Pure Q learners reproduce the tragedy of the commons in every climate scenario.
- Moderate enforcement at p detect = 0.3 saves cooperation under stable climate. Water rises from 5 to 685.
- The SAME enforcement fails under any climate stress. Even p detect = 0.9 cannot rescue shock.
- Discount factor gamma matters at the margins under stable climate but is dominated by climate dynamics under stress.

FUTURE WORK

- Continuous or adaptive action space so agents can scale extraction with current regeneration.
- Climate aware monitor that tightens the fair share threshold during droughts.
- Deep reinforcement learning extension. Comparison with human commons experiments (Ostrom style).

Thank you